For undirected graphs, this means that in a connected graph, any vertex can be reached by all other vertices, while for undirected graphs, it means that each vertex has at least an in-going or an out-going edge.

For either kind, it means that the number of edges is at least $|V| - 1$.

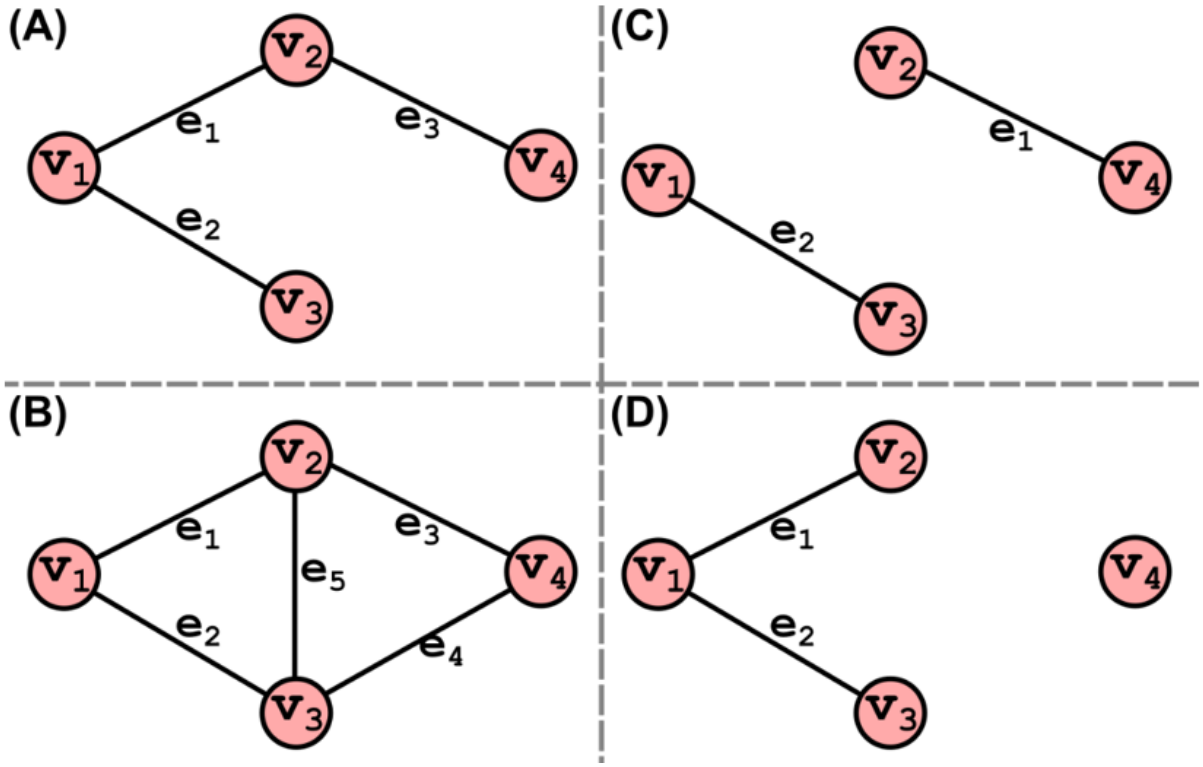


Figure 14.5 Connected graphs ((A), (B)) versus disconnected graphs ((C), (D)).

Figure 14.5 shows a few examples of connected and disconnected undirected graphs; the notion of connected graph makes the most sense, in fact, for undirected graphs, while for directed graphs we introduce, instead, the notion of *strongly connected components*, illustrated in figure 14.6.